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Ionic conductivity of lithium in spinel-type $Li_{4/3}Ti_{5/3}O_4$–$LiMg_{1/2}Ti_{3/2}O_4$ solid-solution system

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ABSTRACT

Spinel type Mg-doped lithium titanium oxides of $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$ has been investigated in terms of the relationship between the crystal structure and the ionic conductivity by both computational and experimental techniques. Reduction of the energy barrier for Li^+ hopping was indicated by the computations based on the first-principles density functional theory (DFT) and long-range Coulombic interactions by doping Mg^{2+} into tetrahedral sites of spinels, including increase of ionic conductivity of Li^+. However, the experimental measurements for activation energies and ionic conductivities of Li^+ were opposite to the expectation from computational study. One of the reasons for the discrepancy was neglecting defect formation energy in the spinel $Li_{4/3}Ti_{5/3}O_4$ in the computations, which was supported by computations assuming Schottky-like Li vacancy formation. In addition, severe increase of activation energy was indicated by doping Mg^{2+} ions at the composition range, $x>0.4$ in experiments. The anomalous increase of activation energy was discussed by adopting percolation theory.

1. Introduction

Rechargeable lithium ion batteries (LIBs) are attracting tremendous interest to overcome the recent growing environmental concerns, such as global warming, as power sources for hybrid electric vehicles (HEVs) and/or electric vehicles (EVs). The key factors of LIBs for these applications are safety, long-life durability, high energy, low cost, and high power.

In this respect, spinel-type $Li_{4/3}Ti_{5/3}O_4$ material has been highlighted as one of the promising anode materials because of its excellent cyclic stability. In addition, $Li_{4/3}Ti_{5/3}O_4$ showed relatively high electrochemical potential at around 1.5 V versus Li metal extrusion potential, so that this material can be cycled safely even at a fast charge/discharge condition [1–4].

Fig. 1(a) shows crystal structure of spinel-type $Li_{4/3}Ti_{5/3}O_4$. It belongs to cubic symmetry with cation distribution $(Li)_{8a}[Li_{1/3}Ti_{5/3}]_{16d}O_4$ (subscripts 8a and 16d indicate the cation site of 1 symmetry). In detail, oxide ions at 32e sites form a cubic closest packed (ccp) structure, tetrahedral 8a sites are occupied by Li ions, and octahedral 16d sites are filled with Li and Ti ions in the molar ratio of 1:5. It is noted that the remaining half of octahedral cation sites in the ccp structure were vacant (16c sites). Soubeyroux et al. have reported the temperature dependence of Li^+ ion distribution in the spinel chlorides by powder neutron diffraction technique [5]. A gradual migration of the involved lithium ions from the 8a site into the 16c vacant site was observed with increasing temperature, so that these 16c-vacancy sites served as migration pathways for Li ions at 8a sites, forming 3-dimensional diffusion network as shown in Fig. 1(b). The same diffusion pathway of 8a–16c–8a mechanism was confirmed in spinel $LiMn_2O_4$ by NMR technique [6] and molecular dynamics simulation [7]. Recently, the 8a–16c–8a Li diffusion pathway was also indicated by first-principle calculation in spinel $LiMn_2O_4$ [8] and $Li_4Ti_5O_{12}$ [9].

Cationic-charge distribution in $Li_{4/3}Ti_{5/3}O_4$ compound is known to be Li^+ and Ti^{4+}, so that *d* orbital of Ti ions is empty. Therefore, this compound is known as pure ionic conductor without electronic conduction, showing very low electronic conductivity of 10^{-13} S cm^{-1} at room temperature [10]. Since such a low electronic conductivity may depress the rate performance of $Li_{4/3}Ti_{5/3}O_4$ electrodes, several attempts were made to increase electronic conductivity. For example, Chen et al. succeeded to improve an electronic conductivity up to 10^{-2} S cm^{-1} by substituting Mg ions partially for Li 8a sites ($Li_{4-x}Mg_xTi_5O_{12}$) to introduce Ti^{3+} (d^1) ions [10]. Therefore, doping strategy would be attractive approach in terms of increasing of an electronic conduction for $Li_{4/3}Ti_{5/3}O_4$ electrodes.

In addition to electronic conductivity, the rate performance of electrodes for LIB was affected by Li ion diffusivity in solid. In this respect, doping strategy may be effective also for the Li^+ diffusivity in the bulk of $Li_{4/3}Ti_{5/3}O_4$ particles. For example, Wakihara et al. reported an improvement of Li^+ diffusivity in spinel $LiMn_2O_4$ by partial replacement of octahedral Mn ions by Co^{3+} or Cr^{3+}, where doped sample showed one order higher diffusion coefficient and improved cyclic performance [11]. However, the studies on Li^+ diffusivity in spinel-type $Li_{4/3}Ti_{5/3}O_4$ related system is limited so far [12,13].

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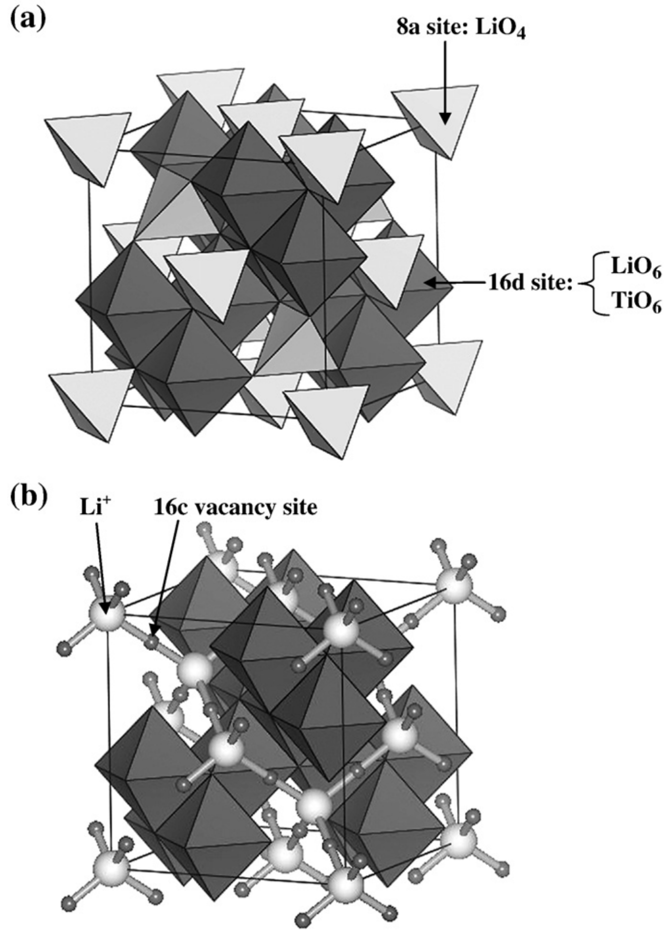

Fig. 1. (a) Crystal structure of spinel-type $Li_{4/3}Ti_{5/3}O_4$. It belongs to cubic symmetry with cation distribution $(Li)_{8a}[Li_{1/3}Ti_{5/3}]_{16d}O_4$ (subscripts 8a and 16d indicate the cation site of $Fd\bar{3}m$ symmetry). Cations at 8a and 16d sites form tetrahedra (brighter) and octahedra (darker) with oxide ion, respectively. (b) Three-dimensional diffusion network of spinel-type $Li_{4/3}Ti_{5/3}O_4$. 16c-vacancy sites served as migration pathways for Li ions at 8a sites.

In the present paper, we have synthesized Mg doped spinel solid solution of $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$ and investigated the relationship between the crystal structure and the ionic conductivity of Li for the sake of systematic study on the Li^+ diffusivity in $Li_{4/3}Ti_{5/3}O_4$ system. It is noted that above solid solution was electronic insulator because of absence of Ti^{3+} ions, so that we adopted a conventional AC impedance technique for the measurement of ionic conductivity. In addition, long-range Coulombic interactions in solid solution were estimated to discuss factors affecting Li^+ diffusivity in the spinels.

2. Models

At first, we calculated the migration pathway of Li and corresponding energy profile for Li hopping between two adjacent 8a sites. All first-principles density functional theory (DFT) calculations were performed by using the Vienna *ab initio* simulation package (VASP) [14,15] with the generalized gradient approximation (GGA-PBE) [16,17] and the projector-augmented wave (PAW) method [18]. An energy cut off was 400 eV and the k-point mesh was chosen such that the number of k-points times the number of atoms in the unit cell was at least more than 5000. In this computation, cationic distribution of $8[(Li^+)_{tet}[Ti^{4+}]_{oct}O_4]^+$ (where subscripts 'tet' and 'oct' indicate tetrahedral and octahedral sites in spinel structure, respectively) was assumed in order to reduce configurational freedom of Li^+/Ti^{4+} arrangement at octahedral 16d sites. The cubic lattice parameter was assigned and held constant, using the values reported in Ref. [19], while the fractional coordinates of ions were allowed to relax. A jellium background was adopted to neutralize the lattice for computations of charged lattice. The nudged elastic band (NEB) method was used to investigate the minimum energy of pathways of the lithium hopping from one lattice position to adjacent sites.

The lowest activation energy for lithium ion jump is obtained for a straightforward migration path between two adjacent 8a sites (Fig. 2), confirming 3-dimensional 8a–16c–8a pathways as depicted in Fig. 1(b). The corresponding energy profile is presented in Fig. 3 as a function of Li coordinates, showing energy maximum at 16c site. Assuming the spinel Li–Ti–O system belongs to classical concept of ionic crystal which is composed of ionic bonds between cations and anions, the energy of system can be described by two interactions: One is long-range Coulombic interactions, and the other one is short-range repulsive interactions owning to electron orbital overlapping between jumping Li^+ and surrounding O^{2-} ions. In particular, latter repulsive interactions are sensitive to interionic distance, such that repulsive energy is sharply increased with shortening the neighboring Li^+–O^{2-} distance. For Li^+ jump between two 8a sites, the minimum interionic distance of Li^+ and O^{2-} (bottle-neck site as shown in Fig. 2) is located around the middle of 8a and 16c sites, while 16c vacant sites offer largest spatial cavity for Li^+ ions. Hence, the short-range repulsive energy is considered to be maximized and minimized at bottleneck and 16c sites, respectively. Since the energy maximum lies not at bottle-neck sites but at 16c sites, the contribution of short-range repulsive interactions would be relatively smaller than that of Coulombic interactions in spinel Li–Ti–O system.

For the simulations of the energy profiles for Mg doped Li–Ti–O spinel system, systematic study by first-principles DFT approach is difficult due to configurational freedom for $Li^+/Mg^{2+}/Ti^{4+}$ arrangement at cation sites. Instead, we calculated the variation of Coulombic energies for Li^+ jump using averaged charge distribution at cation sites with Ewald method. In detail, the structural model was referred from Fig. 2 and Ref. [20] in which Mg^{2+} ions selectively occupied tetrahedral sites. (We discussed and confirmed the cationic distribution later.) Therefore, average charge was assigned for the tetrahedral and octahedral sites represented as $+(1+x)$ and $+(7-x)/2$ in $(Li_{1-x}Mg_x)_{tet}[Li_{(1+x)/3}Ti_{(5-x)/3}]_{oct}O_4$, respectively. Note that mobile Li^+ ion is fixed its charge as $+1$. Fig. 4(a) shows obtained energy profiles for Li^+ jump in the spinel with various composition x. The

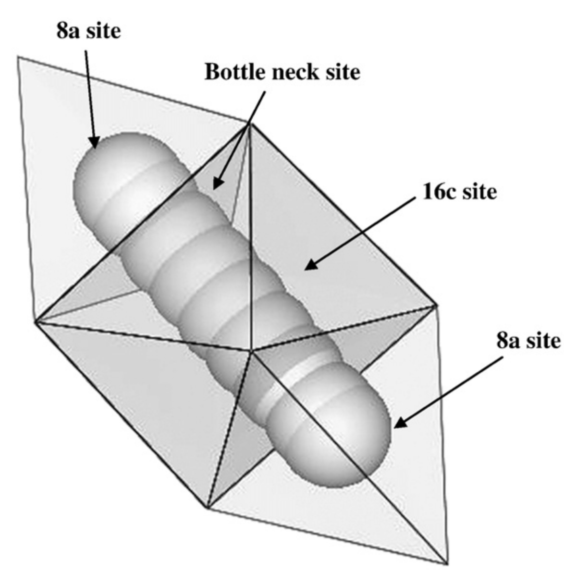

Fig. 2. The calculated trajectory of Li^+ between two adjacent 8a sites in spinel-type $Li_{4/3}Ti_{5/3}O_4$.

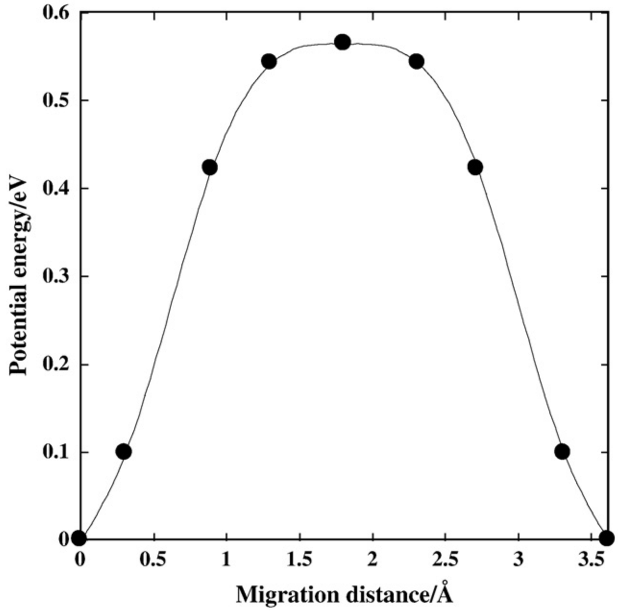

Fig. 3. Calculated energy profile of Li^+ jump by first-principles DFT in the $Li_{4/3}Ti_{5/3}O_4$.

energy maximum lies at 16c sites for all the composition, showing qualitative accordance with first-principles results shown in Fig. 3. Therefore, the Coulombic interaction plays an important role for the ionic conduction in terms of energy barrier during site hopping. In addition, we have calculated the other conceivable pathway that pass through tetrahedral vacancy sites (48f sites) and octahedral 16c vacancies, namely 8a–16c–48f–16c–8a path, by Coulombic calculations, and the results showed much higher activation energy than 8a–16c–8a path (see Fig. 4(b)). Therefore, it was confirmed our assumption of 8a–16c–8a hopping mechanism. The energy difference between maximum and minimum ones, or energy barrier, is summarized at Fig. 4(b) as a function of composition x, and it decreased with the amount of Mg doping. Accordingly, the Mg doping into 8a tetrahedral sites would reduce the activation energy for Li^+ conduction in Li–Ti–O spinel system from the calculations of long-range Coulombic interactions.

3. Experimental

The solid solution samples, $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$ were prepared by a conventional solid-state reaction. Stoichiometric mixture of reagents, Li_2CO_3 (99.99%, Kojundo Chemical Lab. Co, Ltd.), $MgCO_3$ (99.9%, Soekawa Chemicals Industries, Ltd.), and anatase-TiO_2 (99.98%, Soekawa Chemicals Industries, Ltd.), were used as starting materials. The mixture was calcined at 700 °C for 24 h on a Pt foil, and then heated at 900 °C for 24 h with intermittent grounding. The obtained powder was pelletized at 1500 kg/cm^2, and sintered at 1000 °C for 72 h.

Some of the pellets were crushed and grounded into powder and subjected to the phase identification and Rietveld refinement of structural parameters by powder X-ray diffraction (XRD) measurement using X'pertProAlpha-1 system (PANalytical). Rietveld refinement of structural parameters was carried out by using RIETAN-2000 [21]. The ionic conductivity measurement was performed by AC-impedance technique using multi-channel VMP3 electrochemical

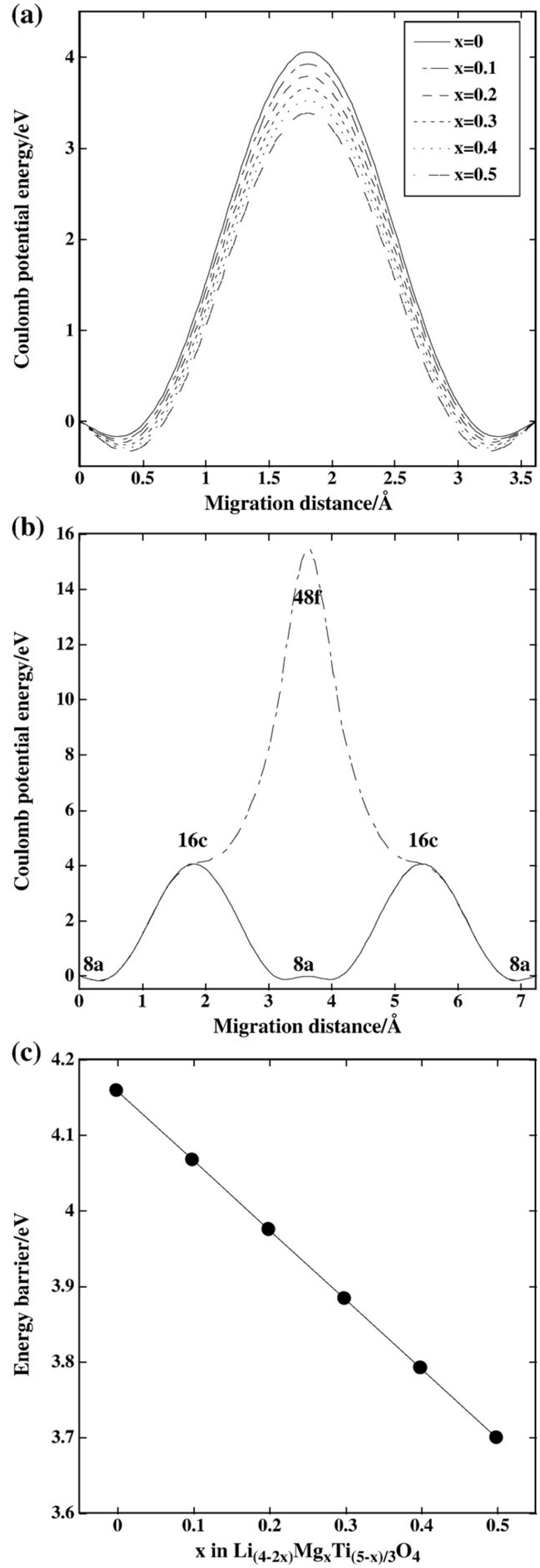

Fig. 4. (a) Calculated Coulombic energy profiles for Li^+ jump in the spinel $(Li_{1-x}Mg_x)_{tet}[Li_{(1+x)/3}Ti_{(5-x)/3}]_{oct}O_4$. Average charges were assigned for the tetrahedral and octahedral sites represented as $+(1+x)$ and $+(7-x)/2$, respectively, and (b) Comparison of calculated energy profiles between 8a–16c–48f–16c–8a and 8a–16c–8a mechanism. (c) The energy difference between maximum and minimum ones, or energy barrier, as a function of composition x.

interface (BioLoig). Both ends of cylindrical sintered pellets were polished up with emery paper, and gold was sputtered to form two-electrode-type specimen. The samples were set in a thermostat and measured the impedance in the frequency range from 5 Hz to 500 kHz and the temperature range from 40 °C to 200 °C.

4. Results

4.1. Crystal structure

Fig. 5 shows powder XRD pattern of obtained solid-solutions, $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$. In addition to the main peaks of spinel phases, weak peaks due to impurity phase of TiO_2 (rutile) was observed as reported previously by Izquierdo et al. [22]. The TiO_2 formation would be ascribed to the partial lithium vaporization at 1000 °C. In detail, the peak position was hardly changed with composition x, indicating small change in the cubic lattice parameter. At the compositional range of $x>\sim0.3$, one can see additional peaks corresponding to the superstructure-type spinel (e.g. $P4_332$) which owes to ordering of the Li^+/Ti^{4+} in octahedral sites. Note that the weak (220) peak around $2\theta\sim30°$ grew its intensity with Mg doping. Since the (220) peak stems from only by X-ray diffraction at tetrahedral cation sites in spinel structure, so that it evidences the replacement of tetrahedral Li^+ by Mg^{2+} with larger scattering factors. Such structural features agreed with previous reports by Dalton et al. [20] for $LiMg_{0.5}Ti_{1.5}O_4$. Summing Fig. 5 up, all the diffraction patterns are able to be indexed by considering three phases, spinel, superstructure-type spinel, and rutile TiO_2.

Structure refinement was carried out with Rietveld analysis by taking account of above three phases, and the lowest reliability factors were obtained as listed in Table 1. Fig. 6 shows the final observed, calculated, and difference profiles for $x=0$ and $x=0.5$ samples. Note that we assumed several constrains for the refinement, such that site occupancies are fulfilled their stoichiometric compositions of $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$, and the Debye–Waller factors for light elements of Li^+ and Mg^{2+} are 1.0. The reliability factor, R_{wp}, and goodness of fit, S, showed less than 12% and 1.8, respectively, indicating the listed structural parameters were conversed sufficiently (Tables 2–4). Since two spinel-phases belonging to $Fd\bar{3}m$ and $P4_332$ symmetry were coexisted at the composition of $x=0.3$, phase transition was occurred

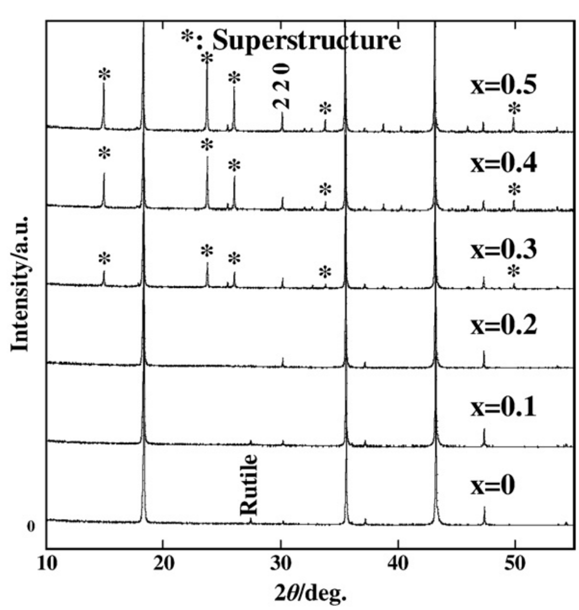

Fig. 5. Powder XRD patterns of obtained solid-solutions, $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$. Additional peaks marked by the asterisks belong to superstructure-type spinel (e.g. $P4_332$).

Table 1
The results of crystal structure analysis by using Rietveld analysis in $Li_{(4-2x)/3}Mg_xTi_{(5-x)/3}O_4$ ($0\le x\le0.5$).

Composition	x					
	0	0.1	0.2	0.3	0.4	0.5
R_{wp}/%	10.91	10.60	10.48	11.18	11.08	11.06
R_p/%	8.33	8.17	7.94	8.58	8.10	8.26
R_e/%	6.46	6.44	6.27	6.42	6.47	6.35
$S=R_{wp}/R_e$	1.69	1.65	1.67	1.74	1.71	1.74

at $x=\sim0.3$ from $Fd\bar{3}m$ to $P4_332$ owing to octahedral cation Li^+/Ti^{4+} ordering. In both phases, the majority of Mg and all the Ti ions are selectively distributed at tetrahedral and octahedral cation sites, respectively, and remaining sites are occupied by Li^+ ions. For the superstructure phase ($x>0.3$), Ti^{4+} ions never placed at octahedral 4b sites, but at octahedral 12d sites. These results agree with reported results by Deschanvres et al. [19] and Dalton et al. [20] for the samples at $x=0$ and 0.5. Fig. 7 presents the variation of cubic-lattice parameters of two spinel phases. The figure indicates that the superstructure formation ($P4_332$) around $x\sim0.3$ caused slight expansion of the lattice (~0.1 vol.% expansion from $Fd\bar{3}m$ to $P4_332$). However, the range of volume expansion by Mg doping is fairly small, showing 0.67 vol.% difference between both compositional ends. Therefore, it is expected that the volume change by Mg-doping does not affect to the ionic conductivity of Li in the $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$ system.

4.2. Ionic conductivity

Fig. 8(a), (b) shows typical AC impedance plots for $Li_{4/3}Ti_{5/3}O_4$ (313 K, 333 K and 353 K) and $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$ ($x=0$, 0.1, 0.2

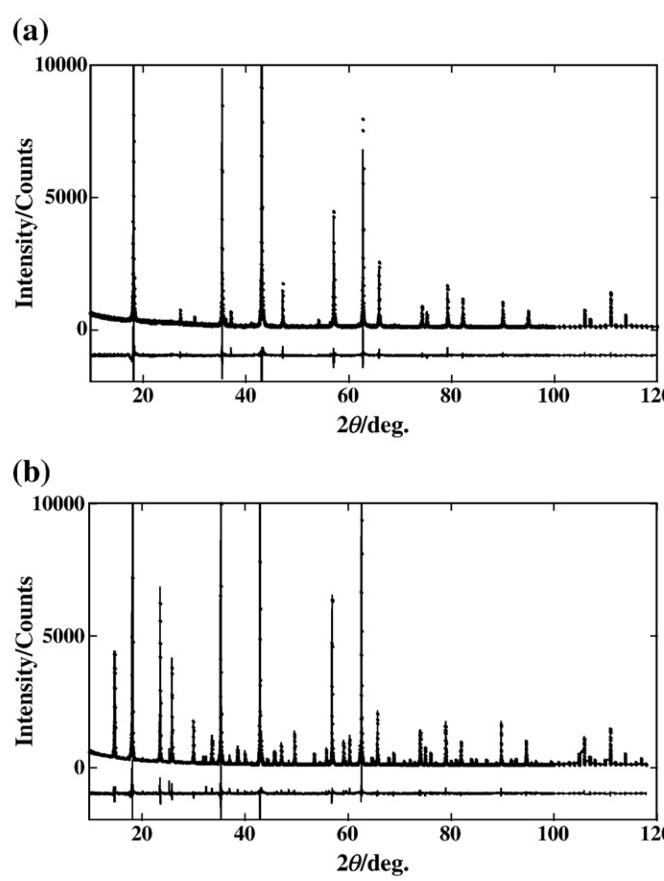

Fig. 6. The final observed, calculated, and difference profiles with Rietveld refinement for composition (a) $x=0$ and (b) $x=0.5$ in $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$.

Table 2
The results of crystal structure analysis by using Rietveld analysis in $Li_{(4-2x)/3}Mg_xTi_{(5-x)/3}O_4$: $Fd\bar{3}m$ ($0 \le x \le 0.5$).

Phase 1: Spinel $Li_{(4-2x)/3}Mg_xTi_{(5-x)/3}O_4$, space group: $Fd\bar{3}m$ (cubic)								
Composition		x	0	0.1	0.2	0.3	0.4	0.5
Lattice parameter		a/Å	8.3620(8)	8.3649(7)	8.3666(9)	8.3703(2)		
Molar fraction/%			95.92	96.39	100	48.20		
8a	Li1	g	1(−)	0.916(−)	0.826(4)	0.74(2)		
	Mg1	g	–	0.084(4)	0.174(−)	0.26(−)		
		x	0	0	0	0		
		y	0	0	0	0		
		z	0	0	0	0		
	Li1	B/Å^2	1(−)	1(−)	1(−)	1(−)		
	Mg1	B/Å^2	–	1(−)	1(−)	1(−)		
16d	Li2	g	0.17(−)	0.176(−)	0.187(−)	0.20(−)		
	Mg2	g	–	0.008(−)	0.013(−)	0.02(−)		
	Ti	g	0.83(−)	0.817(−)	0.8(−)	0.78(−)		
		x	0.625	0.625	0.625	0.625		
		y	0.625	0.625	0.625	0.625		
		z	0.625	0.625	0.625	0.625		
	Li2	B/Å^2	1(−)	1(−)	1(−)	1(−)		
	Mg2	B/Å^2	–	1(−)	1(−)	1(−)		
	Ti	B/Å^2	0.95(2)	0.93(2)	0.91(2)	0.61(9)		
32e	O	g	1	1	1	1		
		x	0.3881(1)	0.3877(1)	0.3873(1)	0.3875(7)		
		y	0.3881(−)	0.3877(−)	0.3873(−)	0.3875(−)		
		z	0.3881(−)	0.3877(−)	0.3873(−)	0.3875(−)		
		B/Å^2	0.97(5)	1.06(4)	1.09(5)	0.8(2)		

at 373 K), respectively. The semicircles at high frequency side were ascribed to the ionic conduction at bulk and grain boundary. Hence, the impedance plots at high frequency region were fitted by two series of RC parallel circuits, and the calculated capacitance for higher frequency sides were around 100 pF, which is typical value of the bulk resistance (for the samples with ~10 mm in diameter and ~1 mm thickness). The calculated bulk conductivities are plotted in Arrhenius manner as shown in Fig. 9. The obtained Arrhenius plots for non-doped $Li_{4/3}Ti_{5/3}O_4$ are consistent with the previous reports by Hayashi et al. [12], Prosini et al. [13] and Takai et al. [23]. Note that the contribution of impurity phase, TiO_2, to the ionic conductivity may be negligibly small, because ionic conductivity was plotted in legalism scale in Fig. 9. The ionic conductivity at 298 K and activation energy calculated by the slope of Arrhenius plots are summarized in Fig. 10 as a function of composition x in $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$. The activation energies and ionic conductivities of samples were increased and decreased, respectively, with doping amount of Mg ions. In detail, slight reduction in the slope for the variation of the activation energy was indicated around the composition $x = 0.3$ (see Fig. 10). This would relate to the phase transition owning to ordering of cation arrangement as indicated in the structure refinements (Fig. 7). However, the magnitude of decrease in activation energy by superstructure formation is apparently smaller than the effect of Mg doping. The activation energy of Li jump in superstructure-type spinel was calculated by *ab initio* DFT. The result showed 0.53 eV, which almost agreed with that in spinel structure (0.53 eV in Fig. 3). Therefore, phase transition would be hardly influenced on the ionic conductivity. In addition, the activation energy showed sharp increase by Mg doping at composition $x > \sim 0.4$, though no structural modification was indicated in this composition range according to our Rietvend refinement. As above, the compositional dependence on activation energies by AC impedance measurements disagreed with our expectations based on the Coulombic-interaction calculation for energy barrier (Section 2). Therefore, the other factors should be affected dominantly for the ionic conduction in Mg-doped $Li_{4/3}Ti_{5/3}O_4$ systems.

5. Discussion

One of the reasons for the discrepancy in the experimental activation-energy behavior from calculated one may be neglecting defect formation energy in the spinel $Li_{4/3}Ti_{5/3}O_4$. In this respect, the

Table 3
The results of crystal structure analysis by using Rietveld analysis in $Li_{(4-2x)/3}Mg_xTi_{(5-x)/3}O_4$: *P42/mnm* ($0 \le x \le 0.5$).

Phase 2: Anatase TiO_2, space group: *P42/mnm* (rectangular)								
Composition		x	0	0.1	0.2	0.3	0.4	0.5
Lattice parameter		$a = b$, c/Å	4.5957(6), 2.9589(−)	4.5944(6), 2.9589(−)				
Molar fraction/%			4.08	3.61				
2a	Ti	g	1(−)	1(−)				
		x	0	0				
		y	0	0				
		z	0	0				
		B/Å^2	1.7(5)	2.2(5)				
4f	O	g	1(−)	1(−)				
		x	0.311(4)	0.298(5)				
		y	0.311(−)	0.298(−)				
		z	0	0				
		B/Å^2	1.3(8)	3(1)				

Table 4
The results of crystal structure analysis by using Rietveld analysis in $Li_{(4-2x)/3}Mg_xTi_{(5-x)/3}O_4$: $P4_332$ ($0 \le x \le 0.5$).

Phase 3: Superstructured spinel $Li_{(4-2x)/3}Mg_xTi_{(5-x)/3}O_4$: space group: $P4_332$ (cubic)								
Composition		x	0	0.1	0.2	0.3	0.4	0.5
Lattice parameter		a/Å				8.3731(1)	8.3792(9)	8.3806(9)
Molar fraction/%						51.80	100	100
8c	Li1	g				0.8(−)	0.61(−)	0.55(−)
	Mg1	g				0.2(−)	0.39(−)	0.45(−)
		x				0	0	0
		y				0	0	0
		z				0	0	0
	Li1	B/Å^2				1(−)	1(−)	1(−)
	Mg1	B/Å^2				1(−)	1(−)	1(−)
4b	Li2	g				0.8(1)	0.92(2)	0.89(2)
	Mg2	g				0(−)	0.01(−)	0.11(−)
	Ti1	g				0.1(−)	0.07(−)	0(−)
		x				0.625	0.625	0.625
		y				0.625	0.625	0.625
		z				0.625	0.625	0.625
	Li2	B/Å^2				1(−)	1(−)	1(−)
	Mg2	B/Å^2				1(−)	1(−)	1(−)
	Ti	B/Å^2				0.72(−)	0.79(−)	0.76(−)
12d	Ti2	g				1(−)	1(−)	1(−)
		x				0.125	0.125	0.125
		y				0.3694(2)	0.3695(1)	0.3689(1)
		z				0.8806(−)	0.8805(−)	0.8812(−)
		B/Å^2				0.72(8)	0.79(2)	0.76(2)
8c	O1	g				1(−)	1(−)	1(−)
		x				0.3905(8)	0.3893(3)	0.3888(3)
		y				0.3905(−)	0.3893(−)	0.3888(−)
		z				0.3905(−)	0.3893(−)	0.3888(−)
		B/Å^2				0.5(2)	0.66(5)	0.78(5)
24e	O2	g				1(−)	1(−)	1(−)
		x				0.1242(8)	0.1275(3)	0.1225(3)
		y				0.1454(9)	0.1430(4)	0.1457(3)
		z				0.8588(7)	0.8592(3)	0.8588(3)
		B/Å^2				0.5(−)	0.66(−)	0.78(−)

In composition $x = 0.3$, we assumed that the compositions of Phases 1 and 3 were both equivalent expressed as $Li_{3.4/3}Mg_{0.3}Ti_{4.7/3}O_4$.

conceivable defect that contribute to the Li conduction is the Li vacancy formation by Schottky-like defect reaction,

$$Li_4Ti_5O_{12} \rightarrow Li_{(4-2y)}Ti_5O_{12-y} + yLi_2O,$$

where two Li vacancy sites formed accompanying one oxygen-site vacancy. To estimate above contribution, we calculated vacancy formation energy by using Coulombic interaction formalism along with the method in Section 2. Direct comparison of lattice energy between spinel and Li_2O is difficult, because empirical parameters for short-range repulsive interactions are required both for spinel and Li_2O. Hence, we set the vacancy formation energy as zero for non-Mg doped sample. (Note that we assume negligibly small difference in repulsive interactions in spinel phase by Mg doping due to small lattice parameter change.) Fig. 11 shows the results of calculated on defect formation energy, which agrees with the experimental tendency of increase in activation energy with composition x. In addition, the magnitude of compositional variation in vacancy formation energy is much larger than that in the migration energy barrier in Fig. 4, where both energies were obtained by consideration of Coulombic interaction solely. Note that the calculations for Fig. 11 did not take into account the phase transition at $x \sim 0.3$, since the energy difference is quite small as mentioned above.

Therefore, the increase in the activation energy by Mg-doping could be ascribed to the vacancy formation energy at tetrahedral Li ion sites. However, for the usage on anode of LIB, defect formation by electrochemical reaction is expected due to electrochemical Li insertion or removal. Accordingly, the improvement of diffusivity could occur for Mg doped $Li_{4/3}Ti_{5/3}O_4$ spinel oxides.

The last part of our discussion deals with the sudden decrease of the ionic conductivity observed at the composition range, $\sim 0.4 < x \le 0.5$. We assume that this conductivity-behavior is related to the percolation-theory [24]. Since Mg ions reside at the tetrahedral cation sites that consist of diffusion-pathways-network, ionic conductivity would reduce proportionally with composition x in the dilute regime of Mg doping. However, the net amount of diffusive Li

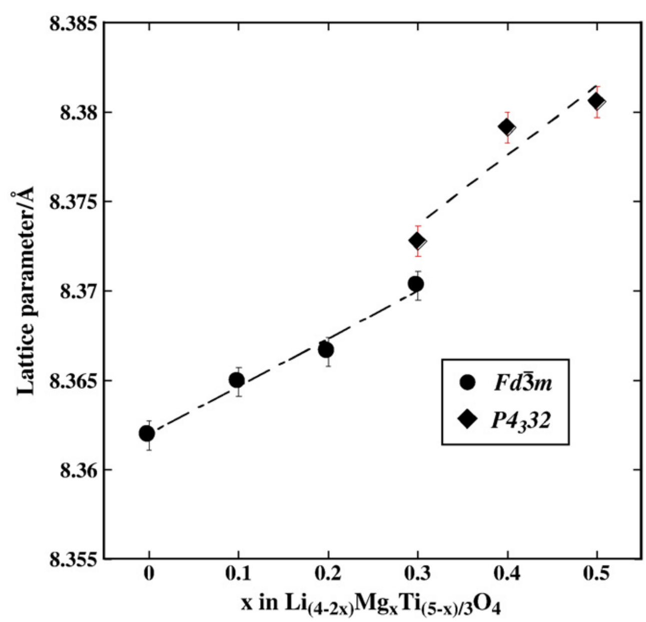

Fig. 7. Variation of cubic-lattice parameters of two spinel phases, $Fd\bar{3}m$ and $P4_332$ as a function of composition x in $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$.

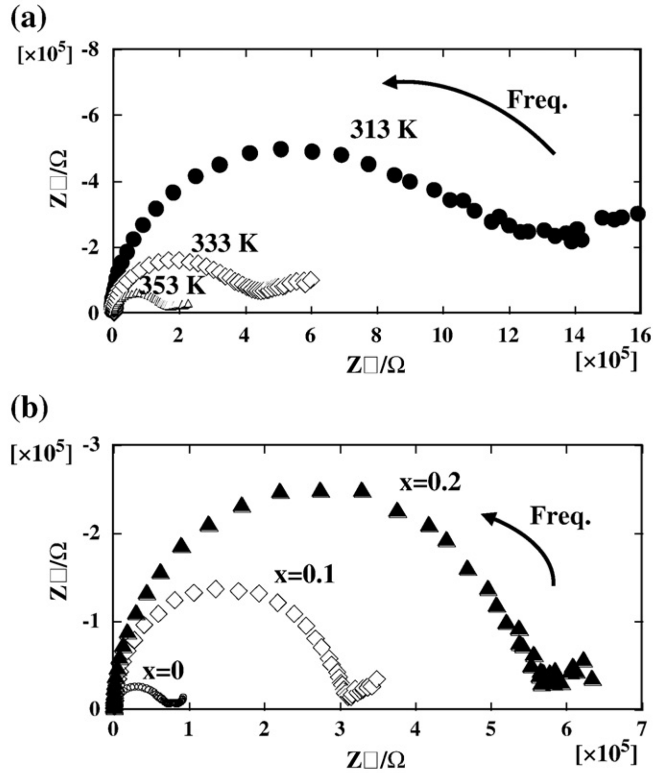

Fig. 8. AC impedance plots of (a) $Li_{4/3}Ti_{5/3}O_4$ (313 K, 333 K and 353 K), (b) $Li_{1-x}Mg_x[Li_{(1+x)/3}Ti_{(5-x)/3}]O_4$ ($x=0$, 0.1, 0.2 at 373 K). The semicircles at high frequency side were ascribed to the ionic conduction at bulk and grain boundary.

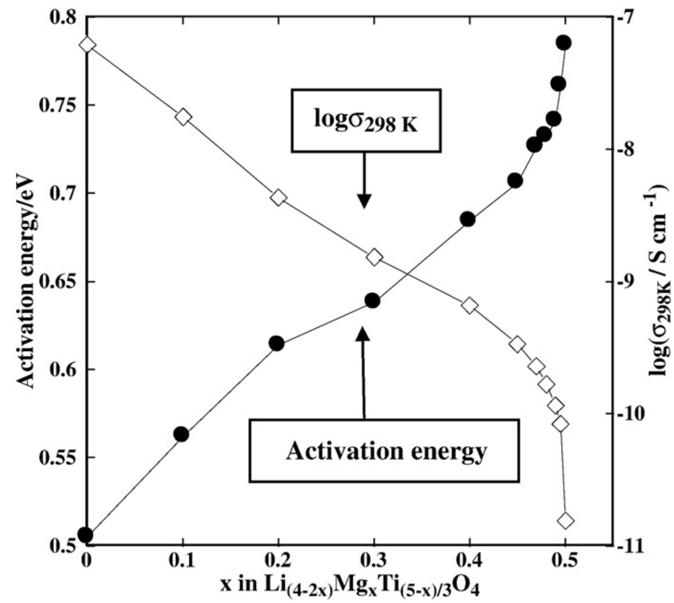

Fig. 10. The ionic conductivity at 298 K and activation energy calculated by the slope of Arrhenius plots as a function of composition x in $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$.

Inaguma et al. succeeded to account for the ionic conduction behavior observed in (Li,Na,La)TiO_3 perovskite materials by the percolation theory. [25–28]. We calculate the net amount of conductive Li ions belonging to the largest cluster as a function of Mg composition x assuming 8a–16c–8a diffusion pathway (known as 'diamond lattice') [24]. In detail, Li and Mg ions were randomly placed at 8a sites and counted the size of Li cluster connected by neighboring Li–Li ions [24].

ions is reduced less than proportional manner at intermediate regime of Mg concentration, because part of Li ions at tetrahedral sites is surrounded by Mg ions. As a result, the network of diffusion-pathway is cut by Mg despite the existence of Li ions at tetrahedral site at a certain amount of Mg ions (known as percolation-threshold). In fact,

Fig. 12 presents the variation of the net amount of conductive Li ions versus composition x. For comparison purpose, that of nominal amount of Li ions is displayed as hatched line in the figure. As seen in the figure, the net amount of the conductive Li ions became almost zero at the composition $x > \sim 0.57$ (or Li concentration at tetrahedral

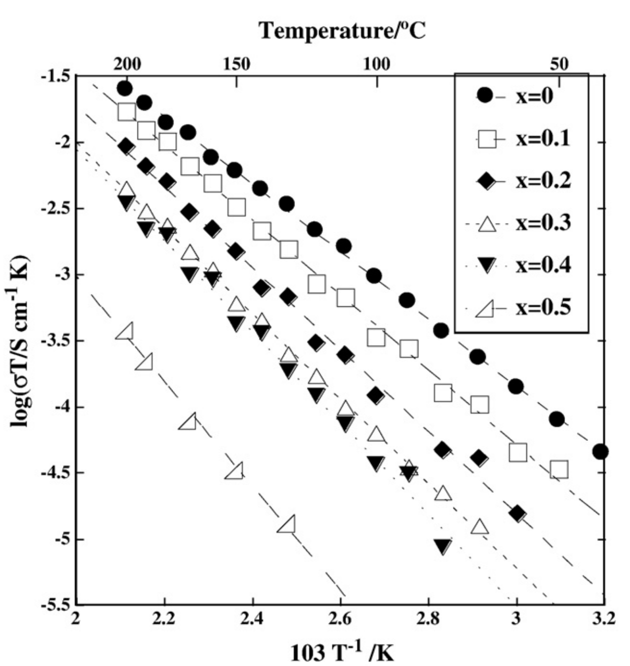

Fig. 9. Temperature dependence of ionic conductivity in $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$ ($0 \leq x \leq 0.5$).

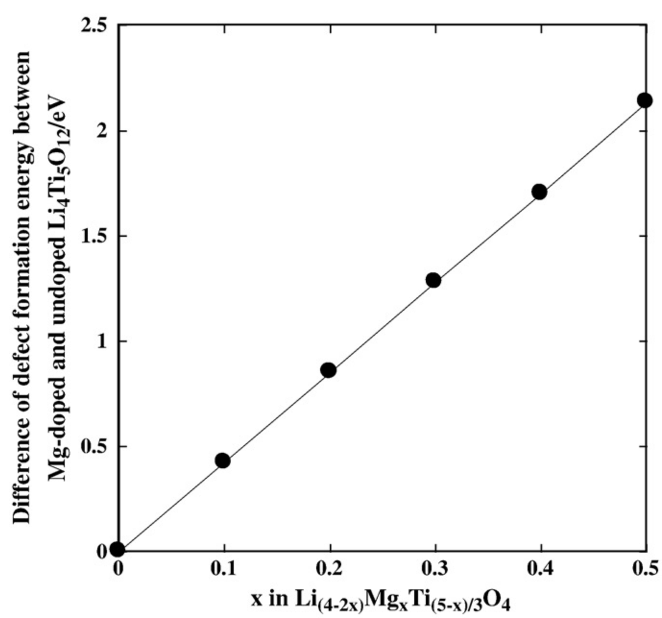

Fig. 11. The calculated results on defect formation energy as a function of composition x in $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$. We set the vacancy formation energy as zero for non-Mg doped sample.

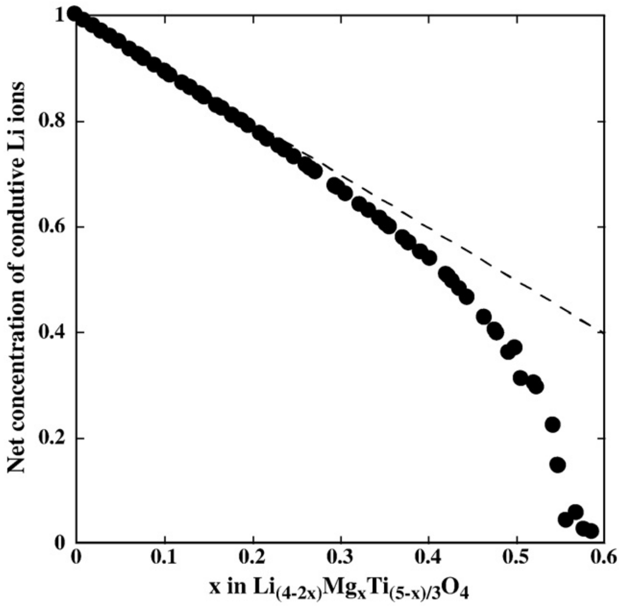

Fig. 12. Variation of net amount of Li ions which consist of the largest cluster connected by nearest neighboring Li ions (solid symbol). Hatched line indicates nominal concentration of Li ions at tetrahedral sites of the spinel.

site, $1-x$, is less than ~0.43), which agrees with known percolation threshold for diamond network ($p_c=0.4301$). One can see rapid decrease of the net amount of conductive Li ions at composition x~>0.45, which correspond to the observed conductivity behavior mentioned above. Thus, we suggest that rapid decrease of ionic conductivity observed at composition x>~0.45 are related to the idea of percolation theory. Note that one can see also an increase of activation energy at composition x>~0.45 in Fig. 10, though the percolation driven decrease of ionic conductivity would not affect activation energy. We expect the change in activation energy is due to the ionic conduction using conductive pathway other than 8a–16c–8a connection in $Fd\bar{3}m$ symmetry. To account for the conductivity phenomena around x>~0.45, further quantitative investigations are needed, such as kinetic-Monte Carlo simulation.

6. Conclusions

Present computational and experimental investigations on $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$ revealed several structural factors that affect Li^+ diffusion in spinel type Li–Ti–O system. By using first-principles DFT calculations, we revealed 3-dimensional 8a–16c–8a diffusion pathway in spinel and significance of long-range Coulombic interactions rather than short-range repulsive interactions. In terms of energy barrier for Li^+ hopping, divalent Mg^{2+} doping enhances the ionic conductivity of lithium by the calculations on long-range Coulombic interactions. In contrast, the calculations based on Coulombic interactions also predict that ionic conductivity would be depressed by Mg^{2+} doping in terms of Li^+ vacancy formation energy assuming Schottky-like mechanism. The experimental results indicated that latter contribution of Li^+ vacancy formation is crucial for the system of spinel Li–Ti–O.

In addition, we confirmed that the phase transition (from $Fd\bar{3}m$ to $P4_332$) due to octahedral Li^+/Ti^{4+} ordering has negligibly small impact to the ionic conductivity behavior. Percolation-theory driven decrease of ionic conductivity was significant and indicated at the composition x>~0.45 in $Li_{(4-2x)}Mg_xTi_{(5-x)/3}O_4$.

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